PROJECT RELAY

The Autonomous Dispatch Desk

Clear decisions. Safe reservations. Recoverable exceptions.

THE PROBLEM / THE SOLUTION

Manual availability checks and unanswered assignments leave deliveries idle. RELAY gives operators one desk to rank riders, progress work and recover stuck jobs, with a reliable explanation of every decision.

DISPATCHER WORKFLOW

Create > ready > scores / override > assign > confirm > pickup > deliver
Reject or timeout > investigate manual queue > reassign as a new attempt.

EXPLAINABLE RECOMMENDATIONS

Hard gates: available, covers both zones and has enough free capacity.

Lowest cost: priority-weighted zone travel + workload + deadline lateness.

Stable ties and stored breakdowns explain ranks. Another eligible choice requires a reason; hard gates always apply.

AUTOMATIC EXCEPTION HANDLING

Persist expiresAt: default two hours, configurable for a short demo. Startup scans recover overdue work after downtime.

Rejection or expiry atomically enters manual, ends the attempt, releases capacity once and appends history.

Investigate retained attempts, then reassign.

ONE MODULAR MONOLITH / FOUR WORKFLOW COLLECTIONS

React + TypeScript
Vite dispatch desk

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Express + TypeScript
Zod / services / worker

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MongoDB replica set
Mongoose transactions

Jobs track delivery state. **Riders** track availability and reserved workload.
Assignments retain each attempt, receipt and score. **Events** explain the ordered history.

THE IMPORTANT TRADE-OFF

Fast counters require replica-set transactions: state, capacity, attempt and audit commit together. Guarded writes and unique indexes prevent conflicts. Polling avoids extra infrastructure but adds processing latency.

WITH ANOTHER WEEK

1. Authenticate roles and strengthen actor/audit access.
2. Monitor worker lag and counter drift; add reconciliation and deployment safeguards.
3. Improve travel estimates and post-pickup exceptions using observed outcomes.

Local workflow verified with real MongoDB, automated races and Chromium. Demo actors; simulated travel.